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[Paper 2] Paper "DM-Highres" - 1e11q High-Res DM Run

This page presents results from the AGORA DM-only run of a $\sim 1e11$ Ms halo at ~ 100 pc resolution being led by Hahn, Simpson, etc. (Paper "2").

The following page includes consensus and updates as recently as from the Paper Group "2" discussions on Feb. 5, 2016, and from the Paper "2" code leaders discussion on Feb. 18, 2016. Important items are in red color.

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1. Prerequisite for Paper "2" Participants

- (1) AGORA cosmological ICs: [WG III Workspace](#), also described in Section 2.1 of [Flagship paper](#)
- (2) First AGORA DM-only comparison: [Paper "1" Workspace](#), also described in Section 5 of [Flagship paper](#)
- (3) Up-to-date participants list: See the table in the most recent progress report ([here](#) as of Aug. 2016)

2. Overview of the Initial Condition

- (1) A $\sim 10^{11} M_{\text{sun}}$ halo with a quiescent merger history (same target halo as in Flagship Paper, a.k.a. 1e11q)
- (2) MUSIC parameter file of the IC:
 - the [old .conf file](#) used in the Flagship Paper with cuboidal highest resolution region (posted Nov. 20, 2012) is superseded by a [newer .conf file with ellipsoidal highest resolution region](#) (posted Jan. 17, 2014)
 - see [this section](#) in WG III's Workspace for more details e.g. merger tree, and Section 5.1 of [Flagship paper](#)
 - requires an update to MUSIC v1.5 or above (see the [v1.5 release note](#) in [MUSIC Google group](#))
- (3) Change levelmax=12 to 13 to increase particle mass resolution
- (4) Target halo mass measured in a test run at $z=0$ (4096^3 effective resolution, coarse refinement):

$$M_{\text{vir}} \sim 1.8 \times 10^{11} M_{\text{sun}} \text{ (see note 1), or } M_{200} \sim 1.7 \times 10^{11} M_{\text{sun}} \text{ (see note 2)}$$

[note 1] M_{vir} is defined as the mass enclosed in a sphere of mean density contrast of 360 with respect to the background density

[note 2] M_{200} is defined as the mass enclosed in a sphere of mean density contrast of 200 with respect to the critical density; this follows the definition in Power et al. 2003 so it can be used in the estimation in the next sections (see [this link](#) for more details)

3. Overview of the Proposed Simulation

(1) List of Consensus Simulation Parameters

	Paper "1" (formerly known RUN-1)	Paper "2" (formerly known RUN-2)
levelmax in MUSIC parameter	12	13
Mass resolution in the finest level (note 1)	$237267 M_{\text{sun}}/h = 3.38 \times 10^5 M_{\text{sun}}$	$29658 M_{\text{sun}}/h = 4.22 \times 10^4 M_{\text{sun}}$
SPH gravitational softening length	3.22 comoving kpc until $z=9$,	1.14 comoving kpc until $z=9$,

in the highest-resolution region (note 2)	322 proper pc afterwards	114 proper pc afterwards (note 3)
AMR finest resolution cell size	326 comoving pc i.e. maximum refinement level = 11	163 comoving pc (note 4) i.e. maximum refinement level = 12
AMR refinement criterion	particle over-density of 4	particle over-density of 4

[Note 1] Note the units used. pc means pc, not pc/h; M_{sun} means M_{sun} , not M_{sun}/h

[Note 2] We define "gravitational softening length" as " eps_{grav} " = "equivalent Plummer softening length" = 114 proper pc after $z=9$. This equates to the parameters "Softening[Gas/Halo/Disk/Stars]" in GADGET/GEAR/GIZMO. Note that this is different from the typical definition of "spline size h ", beyond which the force becomes exactly Newtonian (h as defined in [Eq. 4 of Springel et al. 2005](#)). Then, for GADGET/GEAR/GIZMO, $h = 2.8 * \text{eps}_{\text{grav}} = 319$ pc. Note also that the CHANGA/GASOLINE parameter "dSoft" is *not* the equivalent Plummer softening length, *but* $h/2$ in the Springel et al. 2005 definition of h (but keep in mind that dSoft is indeed equal to h defined in some other papers, e.g. [Eq. 14 of Katz et al. 1996](#) or [Eq. 21 of the original Monaghan & Lattanzio 1985](#)). Therefore, to follow the above consensus, dSoft should be set to = 319 pc / 2 = 160 proper pc after $z=9$ (1.60 comoving kpc until $z=9$).

[Note 3] $0.5 * 4 * R_{200} / \sqrt{N_{200}}$, from Mike Kuhlen's suggestion: 50% of $\text{epsilon}_{\text{opt}}$ found in Eq.(15) of [Power et al. 2003](#)
 $= 0.5 * 4 * (1.7e11 * 2e33)^{1/3} * [200 * (4 * 3.14 / 3) * 1.88e-29 * (0.702)^2]^{-1/3} / 3.086e18 / (1.7e11 / 4.22e4)^{1/2} = 114$ pc
 Therefore,
 -> $\text{eps}_{\text{grav}} = 114$ pc for the highest-resolution region
 -> $\text{eps}_{\text{grav}} = 114 \text{ pc} * 8^{[(\text{level}_{\text{max}} - \text{level}_{\text{here}}) / 2]}$ for the lower resolution regions because epsilon goes as $\sim 1/\sqrt{N} \sim \sqrt{m_{\text{particle}}}$

[Note 4] $60 \text{ Mpc}/h / 2^7 / 2^{12} = 163$ pc, where level = 0 is the root grid with 2^7 cells, i.e. levelmin = 7 in the MUSIC parameter

(2) List of Output Redshifts

We recommend total 201 timesteps. The first 21 timesteps equally spaced in redshift, $\Delta z = 0.5$, from $z=20$ to $z=10$, and the remaining 180 timesteps equally spaced in $\log(\text{expansion parameter})$, $\Delta \log(a) = |\log(1/11)|/180$, from $z=10$ to $z=0$. Plus redshift snapshots at $z = 6.0, 3.0, 2.0, 1.0, 0.5, 0.2, 0.0$ at the very least. Thanks to Junhwan Choi, the [list of exact output redshifts can be downloaded here](#).

(3) Simulation Strategy

(3-1) First run the proposed ~ 100 pc resolution simulation down to $z=0$. **Submit the first dataset (snapshot at $z=0$) to Ji-hoon for initial comparison before the next telecon in the week of Mar. 7, 2016.** Our NERSC repository is a preferred place to collect the dataset, but Ji-hoon can work with any other ways. **If you don't have a user account at NERSC, visit [here](#).**

(3-2) Produce multi-panel images and plots on this Workspace, and iterate until each code group arrives at a parameter configuration that makes its code compatible with others.

(3-3) Perform parameter studies as needed. For example, we may perform force resolution test with AMR codes, at lower resolution if needed (i.e. one AMR code should attempt to over-resolve the initial conditions -- to have more than 1 cell/particle initially -- to agree with tree codes that typically start off with a PM with $2x$ the inter-particle resolution. See e.g. [Klypin+\(2013\)](#).

(3-4) Biweekly telecons amongst code leaders to force ourselves. Publish the result in Paper "2".

4. Pilot Study (Late 2013 - 2015)

A few code groups performed fiducial ~ 100 pc resolution runs with varying ICs (cuboidal or ellipsoidal)... READ MORE IN THE ARCHIVED [OLD PAGE](#).

5. First Round Analysis: Comparison at $z = 0$

Below we post the first batch of compilation figures we agreed to produce first. Please read the Code Notes below carefully before coming up with any conclusion. We also kindly remind you of the [internal policy](#) regarding the disclosure of AGORA simulation results. **The [agora-analysis-](#)**

script repository is to be updated (or **Oliver's repository?**) for the common yt script used to carry out the analysis presented here.

(1) Density Projection

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- Notes: 5(6)-panel density-weighted DM density projection (similar to Figure 3 of Flagship paper).

(2) Density Profile

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- Notes: Composite radial profile of DM density centered on the target halo (similar to Figure 4 of Flagship paper).

(3) Mass Function

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- Notes: Composite subhalo mass function (similar to Figure 6 of Flagship paper but in terms of v_{max}). **Subhalos to be identified by Rockstar (publicly available v0.99, not Rockstar-galaxies beta with multiple-mass reso support) by throwing away lower resolution particles and assuming no contamination around the target halo. See [this link](#) in yt documentation for details.**

(4) More Diagnostics to Come

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(5) Code Notes

CODE	SUBMITTED BY	CODE NOTES / CURRENT STATUS
ART-I	Klypin	• An initial test with levelmax=12? (i.e. RUN-1 parameter in Section 3-(1) above, but with a cuboidal IC, not an ellipsoidal IC; <i>note that ART-I was not part of the Flagship paper in which RUN-1 parameter was used</i>) has been carried out as of Aug. 2014, and stored in agora project directory at NERSC: /global/project/projectdirs/agora/ART_NMSU/NbodyLow/. • Another test with levelmax=13? (still with an ellipsoidal IC?) has been completed by Klypin and the snapshots exist on one of the NASA Ames clusters. • This entry was last updated on Aug. 19, 2016.
ART-II	Feldmann (Semenov?)	• An initial test with levelmax=12 (i.e. RUN-1 parameter in Section 3-(1) above, <i>with an ellipsoidal IC</i>) has been carried out in Aug. 2013 by Leitner in addition to his run for the Flagship paper using a cuboidal IC. • Production-level run with levelmax=13 (i.e. RUN-2 parameter in Section 3-(1) above, with an ellipsoidal IC) is currently underway as of Sep. 2016. • This entry was last updated on Sep. 14, 2016.
CHANGA	Wallace (Quinn)	• An ellipsoidal TIPSy IC was generated by Hahn. • A production-level run with levelmax=13 (i.e. RUN-2 parameter in Section 3-(1) above, with an ellipsoidal IC) has been carried out as of Jun. 2016, and stored in agora project directory at NERSC: /global/project/projectdirs/agora/CHANGA/agora_1e11q_l13_ellipsoid/. Visual comparison of the target halo with the Flagship paper at $z=0$ finds a close resemblance. • A production-level run with levelmax=13 (i.e. RUN-2 parameter in Section 3-(1) above, but with a cuboidal IC, not an ellipsoidal IC) was also carried out in Aug. 2014, and is stored in agora project directory at NERSC: /global/project/projectdirs/agora/CHANGA/agora_1e11q_l13_box/. • Note that the meaning of CHANGA/GASOLINE parameter "dSoft" is different from "SPH gravitational softening length" defined in Section 3-(1) above. i.e. CHANGA parameter dSoft = 1.60 comoving kpc until $z=9$, and 160 proper pc afterwards (see note 2 in Section 3-(1) above) • This entry was last updated on Jun. 3, 2016.
ENZO	Simpson	• Initial tests have been carried out to follow the size of maximally refined cells in time in AMR runs. • A production-level run with levelmax=13 (i.e. RUN-2 parameter in Section 3-(1) above, with an ellipsoidal IC) has been completed and is in the agora project directory at NERSC: /global/project/projectdirs/agora/Enzo/L13_1e11q/. • This entry was last updated on Apr. 24, 2017.
GADGET3	Choi	• A production-level run with levelmax=13 (i.e. RUN-2 parameter in Section 3-(1) above) has been carried out as of Apr. 2016, and stored in agora project directory at NERSC: /global/project/projectdirs/agora/Gadget-3-UNLV/AGORA_Cosmo_DMOOnly_L13_1e11q_500kpc_split/. • This entry was last updated on May 20, 2016.
GASOLINE	Shen	• An initial test with levelmax=13 (i.e. RUN-1 parameter in Section 3-(1) above, but with a cuboidal IC, not an ellipsoidal IC) has been carried out as of Oct. 2014, and stored in agora project directory at NERSC: /global/project/projectdirs/agora/GASOLINE/agora_1e11q_l13/.

		• This entry was last updated on Feb. 18, 2016.
GIZMO-AFS	Garrison-Kimmel	• Garrison-Kimmel, Onorbe, Hummels, Hopkins discussed and concluded in Mar. 2016 that GIZMO with adaptive force softening would be a worthy addition to the comparison. • Initial tests have been carried out with levelmax=12 (i.e. RUN-1 parameter in Section 3-(1) above; <i>note that GIZMO was not part of the Flagship paper in which RUN-1 parameter was used</i>), during which bugs in adaptive softening routines have been identified and corrected. Various values of minimum softening lengths were tested, finding that profiles and mass functions converge well with small enough minimum softening. • For production runs, 3 strategies will be used to showcase the adaptive force softening scheme (namely, "low-softening" vs. "high-softening") : (Run-A) $\text{eps}_{\text{min}} = \text{eps}_{\text{grav}}$, (Run-B) $\text{eps}_{\text{min}} = 0.2 * \text{eps}_{\text{grav}}$, and (Run-C) $\text{eps}_{\text{min}} \sim 0.001 * \text{eps}_{\text{grav}}$. Maximum softenings for $\text{eps}_{\text{min}} = \text{eps}_{\text{grav}}$ and $\text{eps}_{\text{min}} = 0.2 \text{ eps}_{\text{grav}}$ are set at 1000 times the minimum softening length. • These runs (i.e. RUN-2 parameter in Section 3-(1) above, with an ellipsoidal IC) are being carried out as of Aug. 2016. Snapshots and parameter files for the runs can be found on Edison in /global/project/projectdirs/agora/paper2/GIZMO_AGS/m11_L13/. Run-A with $\text{eps}_{\text{min}} = \text{eps}_{\text{grav}}$ are in the subfolder eps_min_fid/; Run-B with $\text{eps}_{\text{min}} = 0.2 \text{ eps}_{\text{grav}}$ are in eps_min_pt2fid. • Run-C is still ongoing as numerical issues with time-stepping have been identified that cause artificial clustering of particles when the softening is highly adaptive. • This entry was last updated on Oct. 22, 2016.
RAMSES	Hahn	• Initial tests have been carried out. • This entry was last updated on Feb. 18, 2016.

6. Second Round Analysis: Time Series

Rockstar and **consistent tree** are to be run on all 200 snapshots from $z=20$ to $z=0$ for all participating codes. In principle, we could compare all or a selection of columns from the Rockstar catalogs.

(1) Density Projection

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- Notes: Compare density projections at other redshifts, e.g. $z=0$, 1, and higher- z . Directly compare positions and masses of the N most massive subhalos (circles overlaid on each panel).

(2) Density Profile

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- Notes: Compare density profiles at other redshifts.

(3) Mass Function

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- Notes: Compare subhalo mass function at other redshifts, investigate the effect of mass resolution, refinement strategy in AMR, and adaptive force softening in SPH (issue raised by Hahn, Nagamine).

(4) Velocity Distribution Function

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- Notes: Compare velocity distribution function, and some diagnostic of velocity structure at center (issue raised by Mao, Wechsler, Kim, see e.g. Mao et al. 2013, Ling et al. 2010, Pillepich et al. 2014).

(5) Accretion Rate

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- Notes: Growth of the target halo mass as a function of time.

(6) Merger Trees, Assembly Histories

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- Notes: Using the location and merger events of the N most massive halos, we could propose a way to circumvent the timing discrepancy, i.e. select a snapshot for each code -- possibly slightly offset from one another -- that best represents a given epoch for cross-platform comparison (issue

raised in the Flagship Paper).

[\(7\) Resolution](#)

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- Notes: Maximum resolution (or density) as a function of time similar to Figure 7 of [Luo et al. 2015](#). We could propose a way to make AMR and SPH resolutions the most compatible (issue raised by Nagamine).

[\(8\) More Diagnostics to Come](#)

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